6

Radiologic and Nuclear Imaging in Nephrology

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Options for imaging of patients with kidney disease have changed significantly in recent years. Intravenous urography (IVU) is infrequently used and has mostly been replaced by ultrasound, computed tomography (CT), magnetic resonance imaging (MRI), and nuclear medicine scanning. Rapidly changing computer-based data manipulation has resulted in major technologic advances in each of these modalities. Three-dimensional (3D) or even 4D (time-sensitive) image analysis is now available. *Molecular imaging,* which visualizes cellular function using biomarkers, is providing both functional and anatomic information.

The American College of Radiology (ACR) has published Appropriateness Criteria[1] that suggest the best choice of imaging modality to answer the clinical questions while minimizing cost and potential adverse effects, such as contrast-induced adverse events and radiation exposure. Tables 6.1 through 6.3 list relative radiation exposures, first-choice imaging modalities in kidney disease, and risk estimates, respectively. Risks of imaging and cost need to be balanced against benefits.

ULTRASOUND

Ultrasound is relatively inexpensive and provides a rapid way to assess renal location, contour, and size without radiation exposure. Nephrologists are increasingly undertaking bedside ultrasound examination; the practical techniques as well as the appropriate interpretative skills are discussed in Chapters 5 and 97. Portable ultrasound is available and is essential in the pediatric or emergency setting. Obstructing renal calculi can be readily detected, and renal masses can be identified as cystic or solid. In cases of suspected obstruction, the progression or regression of hydronephrosis is readily evaluated. Color Doppler imaging permits assessment of renal vascularity and perfusion. Unlike the other imaging modalities, ultrasound is highly dependent on operator skills. Limitations of ultrasound include lack of an acoustic window, body habitus, and poor patient cooperation.

Kidney Size

The kidney is imaged in the transverse and sagittal planes and is normally 9 to 12 cm in length in adults. Differences in kidney size can be detected with all imaging modalities. Figure. 6.1 shows the common causes of enlarged and shrunken kidneys.

Renal Echo Pattern

The normal kidney cortex is hypoechoic compared with the fat-containing echogenic renal sinus (Fig. 6.2A). The cortical echotexture is defined as isoechoic or hypoechoic compared with the liver or spleen. In children, the renal pyramids are hypoechoic (Fig. 6.2B) and the cortex is characteristically hyperechoic compared with the liver and the spleen. In adults, an increase in cortical echogenicity is a sensitive marker for parenchymal renal disease but is nonspecific (Fig. 6.3). Decreased cortical echogenicity can be found in acute pyelonephritis and acute renal vein thrombosis.

The normal renal contour is smooth, and the cortical mantle should be uniform and slightly thicker toward the poles. Two common benign pseudomasses that can be seen with ultrasound are the dromedary hump and the column of Bertin. The column of Bertin results from bulging of cortical tissue into the medulla; it is seen as a mass with an echotexture similar to that of the cortex, but it is found within the central renal sinus (Fig. 6.4). The renal pelvis and proximal ureter are anechoic. An *extrarenal pelvis* refers to the renal pelvis location outside the renal hilum. The ureter is not identified beyond the pelvis in nonobstructed patients.

Obstruction can be identified by the presence of hydronephrosis (Fig. 6.5). Parenchymal and pelvicalyceal nonobstructing renal calculi as well as ureteral obstructing calculi can be readily detected (Fig. 6.6). The upper ureter also will be dilated if obstruction is distal to the pelviureteral junction (see Fig. 6.5C). False-negative ultrasound examination findings with no hydronephrosis occasionally occur in early obstruction. Obstruction without ureteral dilation also may occur in retroperitoneal fibrosis and in transplanted kidneys as a result of periureteral fibrosis.

Renal Cysts

Cysts can be identified as anechoic lesions and are a frequent coincidental finding during renal imaging. Ultrasound usually readily identifies renal masses as cystic or solid (Figs. 6.7 and 6.8). However, hemorrhagic cysts may be mistakenly called solid because of increased echogenicity. Differentiation of cysts as simple or complex is required to plan intervention.

Simple Cysts

A simple cyst on ultrasound is anechoic, has a thin or imperceptible wall, and demonstrates through-transmission because of the relatively rapid progression of the sound wave through fluid compared with adjacent soft tissue.

Complex Cysts

Complex cysts contain calcifications, septations, and mural nodules. Instead of being anechoic, these masses may contain internal echoes representing hemorrhage, pus, or protein. Complex cysts may be benign or malignant; cyst wall nodularity, septations, and vascularity strongly suggest malignancy. Complex cysts identified by ultrasound require further evaluation by contrast-enhanced CT (or MRI) to identify abnormal contrast enhancement of the cyst wall, mural nodule, or septum, which may indicate malignancy.

TABLE 6.1 Relative Radiation Doses of Imaging Examinations

Examination	Effective Dose (mSv)
Chest: PA radiograph	0.02
Lumbar spine	1.8
KUB abdomen	0.53
CT abdomen	10
CT chest	20–40
PET-CT	25
Ultrasound or MRI	0

CT, Computed tomography; *KUB,* kidney, ureter, bladder (plain film); *MRI,* magnetic resonance imaging; *mSv,* millisieverts; *PA,* posteroanterior; *PET,* positron emission tomography.

TABLE 6.2 Suggested Imaging in Renal Disease

Pathology	First-Choice Imaging
Acute kidney injury, chronic kidney disease	Ultrasound
Hematuria	Ultrasound or CT
Proteinuria, nephrotic syndrome	Ultrasound Multiphase CT urography
Hypertension with normal kidney function	Ultrasound Consider CTA or MRA
Hypertension with impaired kidney function	Ultrasound with Doppler
Kidney infection	Contrast-enhanced CT
Hydronephrosis identified on ultrasound	Nuclear renogram
Retroperitoneal fibrosis	Contrast-enhanced CT
Papillary or cortical necrosis	Contrast-enhanced CT
Renal vein thrombosis	Contrast-enhanced CT
Kidney infarction	Contrast-enhanced CT
Nephrocalcinosis and nephrolithiasis	CT

These recommendations assume availability of all common imaging modalities.
CT, Computed tomography; *CTA,* computed tomographic angiography; *MRA,* magnetic resonance angiography.
Modified from American College of Radiology. Appropriateness criteria: http://www.acr.org/ac.

Bladder

Real-time imaging can be used to evaluate for bladder wall tumors and bladder stones. Color flow Doppler evaluation of the bladder in well-hydrated patients can be used to identify a ureteral jet, produced when peristalsis propels urine into the bladder. The incoming urine has a higher specific gravity relative to the urine already in the bladder (Fig. 6.9). Absence of the ureteral jet can indicate total ureteral obstruction.

Renal Vasculature

Color Doppler investigation of the kidneys provides a detailed evaluation of the renal vascular anatomy. The main renal arteries can be identified in most patients (Fig. 6.10). Power Doppler imaging is a more sensitive indicator of flow, but unlike color Doppler imaging, power Doppler provides no information about flow direction and

TABLE 6.3 Risk Estimates in Diagnostic Imaging

Imaging Risk	Estimated Risk
Cancer from 10 mSv of radiation (1 body CT)[2]	1 in 1000
Contrast-induced nephropathy in patient with reduced GFR[4]	Uncertain but higher with diabetes or hyperuricemia
Nephrogenic systemic fibrosis[4,6]	1 in 25,000 to 1 in 30,000 (depends on gadolinium agent) Higher risk if GFR >30 mL/min
Death from iodine contrast anaphylaxis[5]	1 in 130,000
Death from gadolinium contrast anaphylaxis[6]	1 in 280,000

CT, Computed tomography; *GFR,* glomerular filtration rate; *mSv,* millisieverts.

cannot be used to assess vascular waveforms. However, power Doppler imaging is exquisitely sensitive for detection of renal parenchymal flow and has been used to identify cortical infarction.

RENAL ARTERY DUPLEX SCANNING

The role of gray-scale and color Doppler sonography in evaluating for renal artery stenosis is controversial. The principle is that a narrowing in the artery will cause a velocity change commensurate with the degree of stenosis, as well as a change in the normal renal artery waveform downstream from the lesion. The normal renal artery waveform demonstrates a rapid systolic upstroke and an early systolic peak (Fig. 6.11A). The waveform becomes damped downstream from a stenosis. This consists of a slow systolic acceleration (tardus) and a decreased and rounded systolic peak (parvus) (see Fig. 6.11B). It also results in a decrease in the *resistive index*, defined as the end-diastolic velocity (EDV) subtracted from the peak systolic velocity (PSV) divided by PSV: (PSV − EDV)/PSV. The normal resistive index is 0.70 to 0.72.

The entire length of the renal artery should be examined for the highest velocity signal. The origins of the renal arteries are important to identify because this area is often affected by atherosclerosis, but the arteries are often difficult to visualize because of overlying bowel gas. Within the kidney, medullary branches and cortical branches in the upper, middle, and lower thirds should be included to attempt detection of stenosis in accessory or branch renal arteries.

Proximal and distal criteria exist for diagnosis of significant renal artery stenosis, usually defined as stenosis greater than 60%. The *proximal* criteria detect changes in the Doppler signal at the site of stenosis and provide sensitivities and specificities ranging from, respectively, 0% to 98% and 37% to 98%.[2,3] Technical failure rates are typically 10% to 20%.[4] Renal artery stenosis also may be missed if PSV is low because of poor cardiac output or aortic stenosis. False-positive results can occur when renal artery velocity is increased because of high-flow states, such as hyperthyroidism or vessel tortuosity. The *distal* criteria are related to detection of a tardus-parvus waveform distal to a stenosis; sensitivities and specificities of 66% to 100% and 67% to 94%, respectively, have been reported.[5,6] Technical failure with distal criteria is much lower than with proximal evaluation (<5%). False-negative results can occur from stiff poststenotic vessels, which will decrease the tardus-parvus effect. The tardus-parvus effect also may be a result of aortic stenosis, low cardiac output, or collateral vessels in complete occlusion, giving a false-positive result.

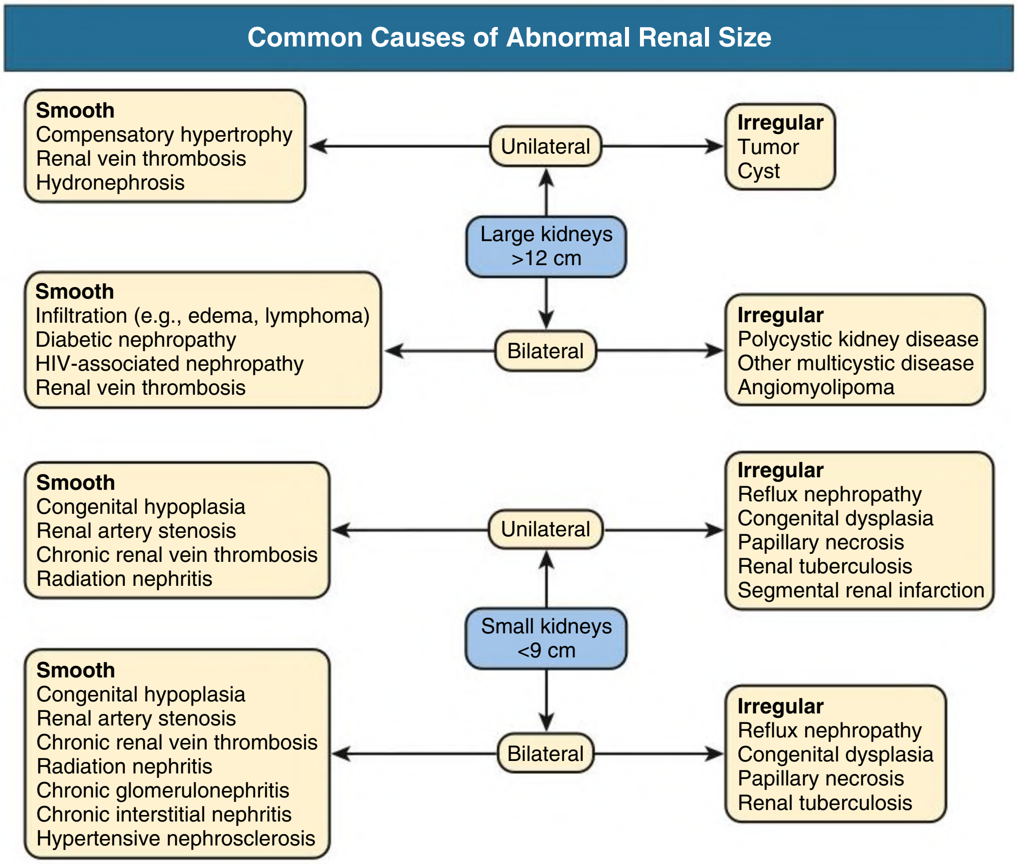

Fig. 6.1 Common causes of abnormal kidney size.

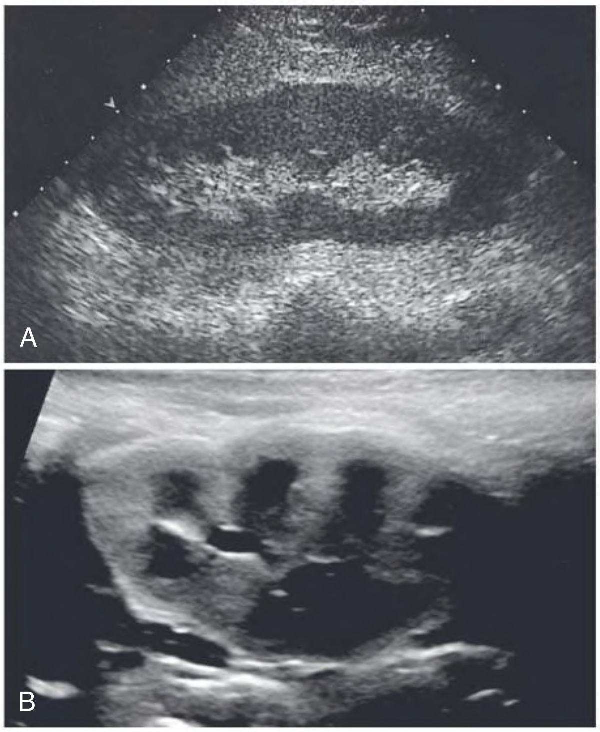

Fig. 6.2 Ultrasound Images of Kidney. (A) Normal sagittal renal ultrasound image. The cortex is hypoechoic compared with the echogenic fat containing the renal sinus. (B) Normal renal ultrasound image in an infant. Note the hypoechoic pyramids.

Fig. 6.3 Nephropathy Associated With Human Immunodeficiency Virus. Enlarged echogenic kidney with lack of corticomedullary distinction. Bipolar length of kidney is 14.2 cm.

Fig. 6.4 Sagittal Kidney Ultrasound. The column of Berlin *(arrows)* is easily identified because of echotexture similar to that of kidney cortex.

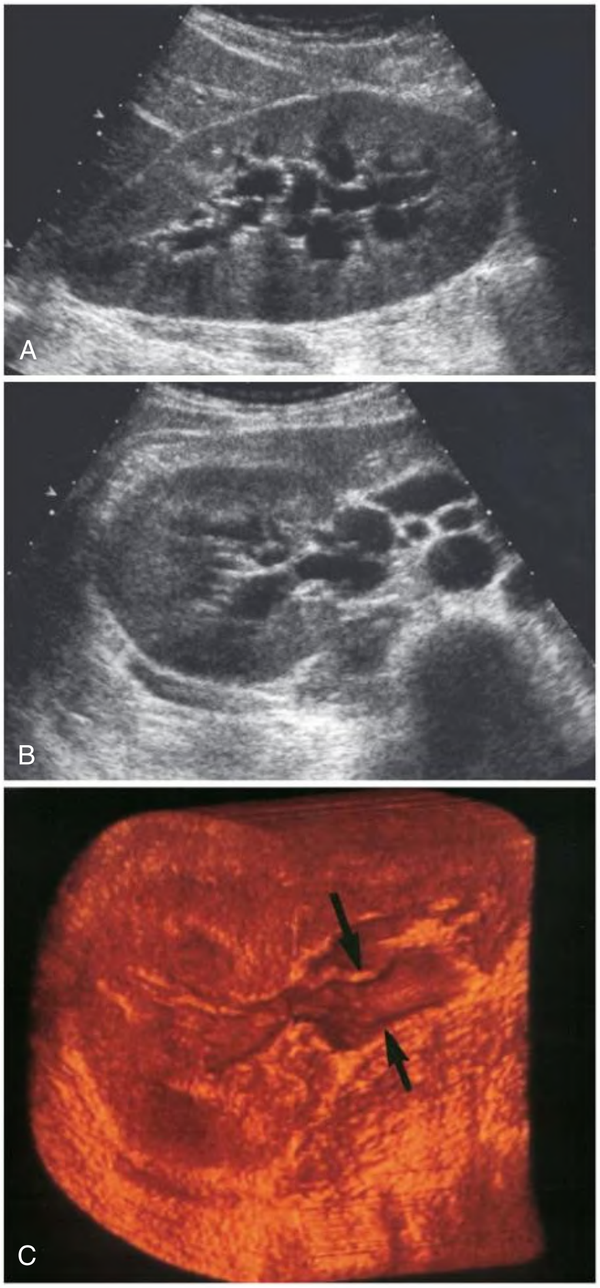

Fig. 6.5 **Kidney Ultrasound Study Demonstrating Hydronephrosis.** (A) Sagittal ultrasound image. (B) Transverse image. (C) Transverse three-dimensional surface-rendered image. *Arrows* indicate the dilated proximal ureter.

Fig. 6.6 **Renal Calculus *(Arrow)* of Upper Pole.** Note the acoustic shadowing *(arrowhead)* on sagittal ultrasound image.

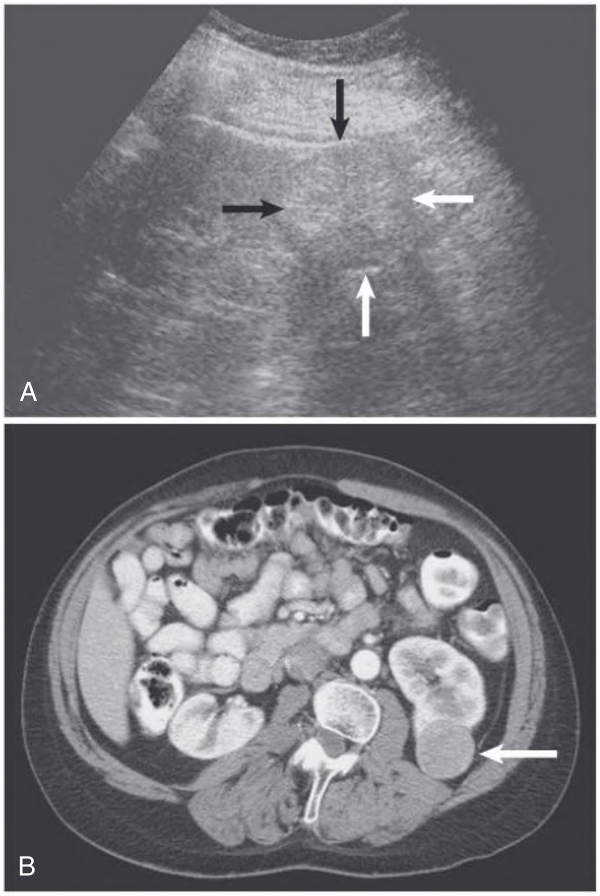

Fig. 6.7 **Evaluation of Kidney Mass.** (A) Sagittal ultrasound image shows large hyperechoic mass arising from the lower pole *(arrows)*. (B) Corresponding contrast-enhanced computed tomography scan shows kidney cell carcinoma *(arrow)*.

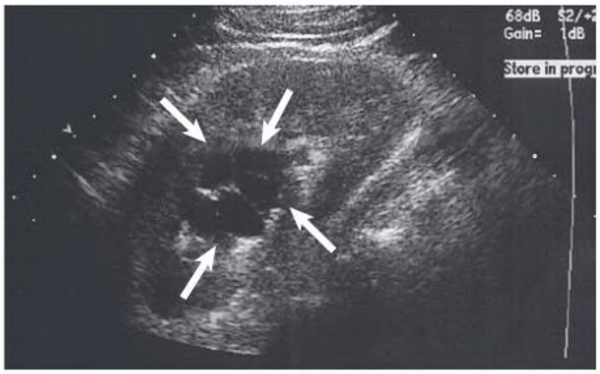

Fig. 6.8 Sagittal ultrasound image of a complex kidney cyst *(arrows)*.

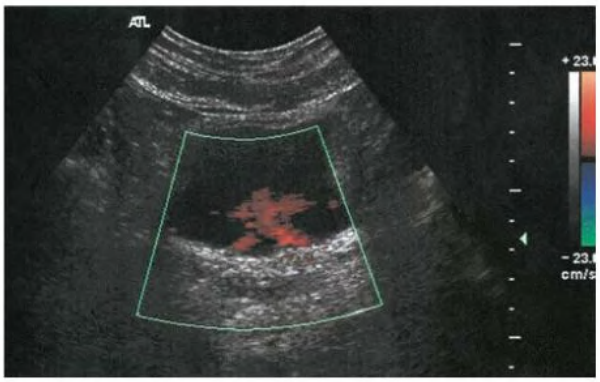

Fig. 6.9 **Bilateral Ureteral Jets in Bladder.** Color Doppler ultrasound study detects this normal appearance.

Fig. 6.10 Transverse color Doppler ultrasound evaluation of the kidney shows the artery as red and the vein as blue.

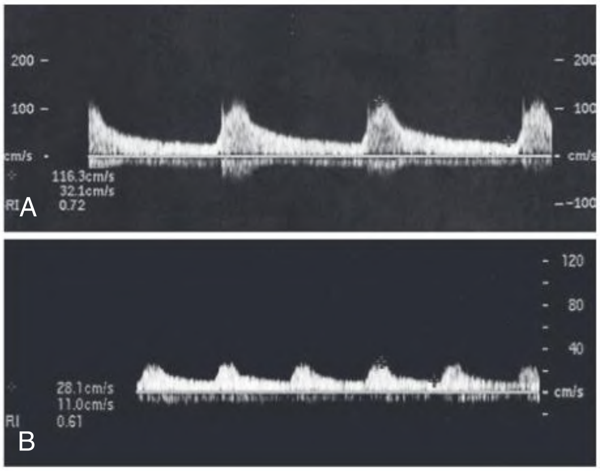

Fig. 6.11 Renal Artery Color Doppler Image and Spectral Imaging. (A) Normal renal artery tracing shows rapid systolic upstroke and early systolic peak velocity (~100 cm/s). (B) Tardus-parvus waveform demonstrates slow systolic upstroke (acceleration) and decreased peak systolic velocity (~20 cm/s) associated with renal artery stenosis. Note different scales on vertical axis.

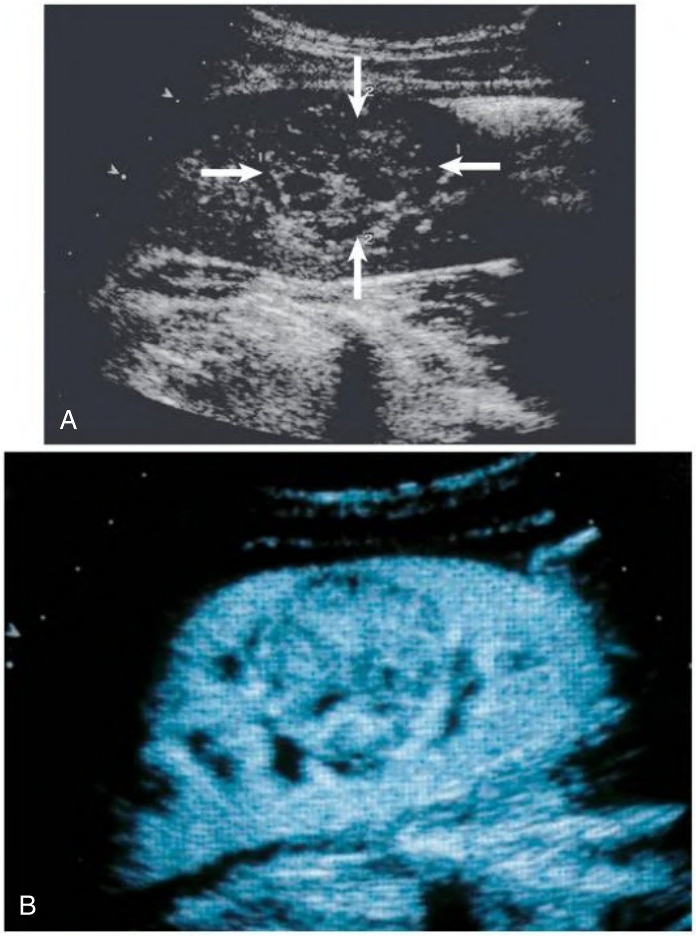

Fig. 6.12 Contrast Ultrasonography. (A) Sagittal kidney ultrasound image with a large, central kidney cell carcinoma *(arrows)*. (B) Central carcinoma better seen after injection of contrast material. (Courtesy Dr. Christoph F. Dietrich.)

Combining the proximal and distal criteria improves detection of stenoses. Sensitivity of 97% and specificity of 98% can be achieved when both the extrarenal and the intrarenal arteries are examined.[7] When it is technically successful, Doppler ultrasound has a negative predictive value of more than 90%.[7] However, reliable results require a skilled and experienced sonographer and a long examination time. Despite these limitations, Doppler studies also have several advantages. Noninvasive, inexpensive, and widely available, Doppler studies also allow structural and functional assessment of the renal arteries and imaging without exposure to radiation or contrast material.

Some physicians prefer computed tomography angiography (CTA) or magnetic resonance angiography (MRA) as a faster and more reliable test than ultrasound, but at present the choice should depend on local expertise and preference. For further discussion of the diagnosis and management of renovascular disease, see Chapter 41.

CONTRAST-ENHANCED AND THREE-DIMENSIONAL ULTRASOUND

Ultrasound contrast agents, initially introduced to assess cardiac perfusion, are now being used to evaluate perfusion to other organs, such as the kidney. These intravenous agents are microbubbles 1 to 4 μm in diameter (smaller than erythrocytes) that consist of a shell surrounding the echo-producing gas core. The microbubbles oscillate in response to the ultrasound beam frequency and give a characteristic increased echo signal on the image. Preliminary studies evaluating renal perfusion in dysfunctional kidneys show reduced flow compared with normal kidneys, as well as improved lesion detection (Fig. 6.12). However, the clinical adoption of microbubble imaging in the kidney remains uncertain, particularly with the general availability and robustness of CT and MRI.

Two-dimensional ultrasound images can be reconstructed into 3D volume images by a process similar to 3D reconstructions for MRI and CT. Potential applications include vascular imaging and fusion with MRI or positron emission tomography (PET).

PLAIN RADIOGRAPHY AND FLUOROSCOPY

Although infrequently used today in high-income countries, contrast urography may still be a key investigation in parts of the world where cross-sectional imaging is not readily available. Plain radiography, often called KUB (kidneys, ureter, bladder), still has an important role in the identification of soft tissue masses, bowel gas patterns, calcifications, and renal location.

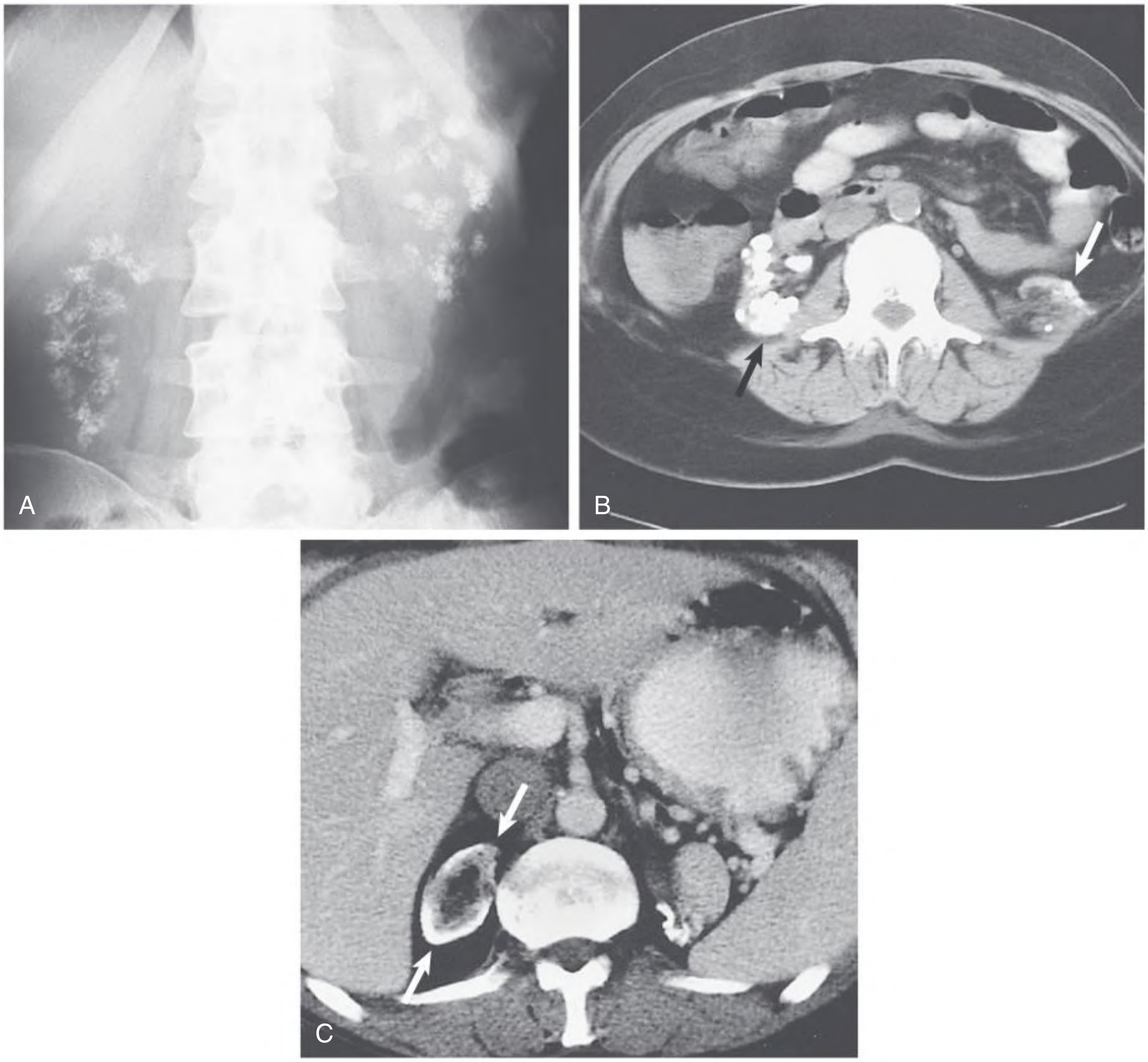

Fig. 6.13 Nephrocalcinosis. (A) Plain radiograph shows bilateral medullary nephrocalcinosis in a patient with distal renal tubular acidosis. (B) Noncontrast computed tomography (CT) scan in a patient with hereditary oxalosis and dense bilateral renal calcification *(arrows).* The left kidney is atrophic. (C) Noncontrast CT scan shows cortical nephrocalcinosis in the right kidney *(arrows)* after cortical necrosis.

Renal Calcification

Most renal calculi are radiodense, although only about 60% of urinary stones detected on CT are visible on plain films.[8] CT demonstrates nonopaque stones, which include uric acid, xanthine, and struvite stones. However, neither CT nor plain radiography may detect calculi associated with protease inhibitor therapy.[9] Oblique films are sometimes obtained to confirm whether a suspicious upper quadrant calcification is renal in origin. CT is the imaging modality of choice for detection of urinary calculi.[10]

Nephrocalcinosis may be medullary (Fig. 6.13A–B) or cortical (Fig. 6.13C) and is localized or diffuse. The causes of nephrocalcinosis are discussed in Chapter 57 (see Box 57.7).

RETROGRADE PYELOGRAPHY

Retrograde pyelography is performed when the ureters are poorly visualized on other imaging studies or when samples of urine must be obtained from the kidney for cytology or culture. Patients who have severe allergies to contrast agents or impaired kidney function can be evaluated with retrograde pyelography. The examination is performed by placing a catheter through the ureteral orifice under cystoscopic guidance and advancing it into the renal pelvis. The catheter is slowly withdrawn under fluoroscopy while radiocontrast is injected (see Figs. 58.2 and 58.11). This provides excellent visualization of the renal pelvis and ureter and can be used for cytologic sampling from suspect areas.

ANTEGRADE PYELOGRAPHY

Antegrade pyelography is performed through a percutaneous renal puncture and is used when retrograde pyelography is not possible. Ureteral pressures can be measured, hydronephrosis evaluated, and ureteral lesions identified (Fig. 58.14). The examination is often performed as a prelude to nephrostomy placement. Both antegrade and retrograde pyelography are invasive and should be performed only when other studies are inadequate.

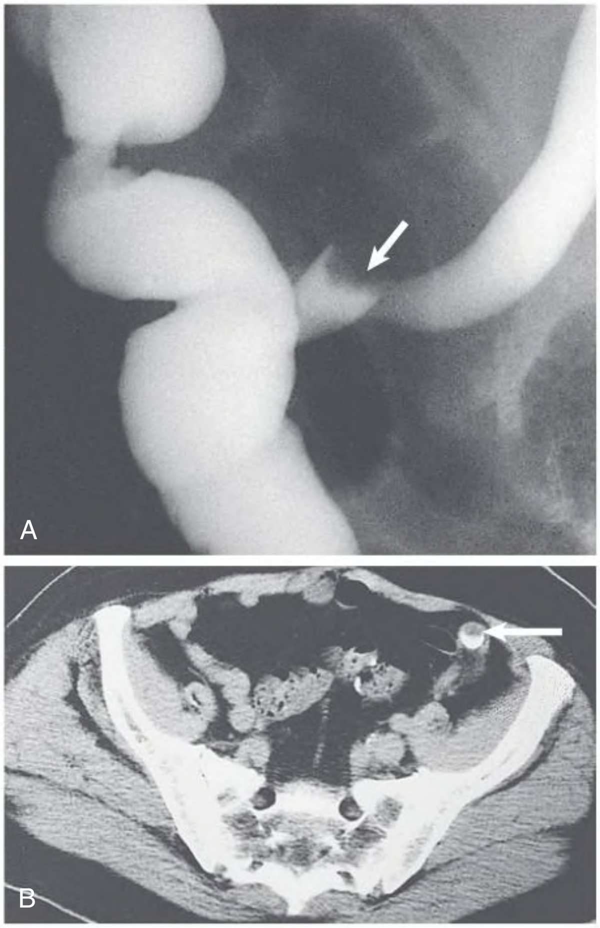

Fig. 6.14 **Imaging of an Ileal Conduit.** (A) Loop-o-gram. A recurrent transitional carcinoma is present in the reimplanted left ureter *(arrow).* (B) Computed tomography scan clearly shows the tumor as a filling defect in anterior aspect of the opacified ureter *(arrow).*

IMAGING ILEAL CONDUITS

After cystectomy or bladder failure, numerous types of continent or incontinent urinary diversions can be surgically created. One of the most common diversions is the ileal conduit: an ileal loop is isolated from the small bowel, and the ureters are implanted into the loop. This end of the loop is closed, and the other end exits through the anterior abdominal wall. This type of conduit can be evaluated by an excretory or a retrograde study. The excretory or *antegrade* study is performed and monitored in the same way as an IVU. A *retrograde* examination, also referred to as a "loop-o-gram," is obtained when the ureters and conduit are suboptimally evaluated on the excretory study. A Foley catheter is placed into the stoma and contrast slowly instilled. The ureters should fill by reflux because the ureteral anastomoses are not of the antireflux variety (see Fig. 6.14).

CYSTOGRAPHY

A cystogram is obtained when more detailed radiographic evaluation of the bladder is required. *Voiding* cystography is performed to identify ureteral reflux and assess bladder function and urethral anatomy. A urethral catheter is placed into the bladder, the urine drained, contrast infused, and the bladder filled under fluoroscopic guidance. Early frontal and oblique films with the patient supine are obtained while the bladder is filling. Ureteroceles are best identified on early films. When the bladder is full, multiple films are obtained with varying degrees of obliquity. Reflux may be seen on these films. To obtain a voiding cystogram, the catheter is removed, the patient voids, and the contrast is followed into the urethra. Occasionally, bladder diverticula are seen only on the voiding films. When the patient has completely voided, a final film is used to assess the amount of residual urine and the mucosal pattern.

Radionuclide cystography is an alternative often used in children. It is useful in the diagnosis of reflux, but it does not provide the detailed anatomy seen with contrast cystography.

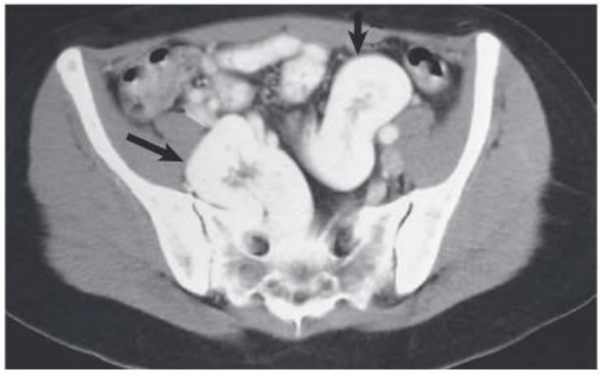

Fig. 6.15 Bilateral pelvic kidneys *(arrows)* on computed tomography.

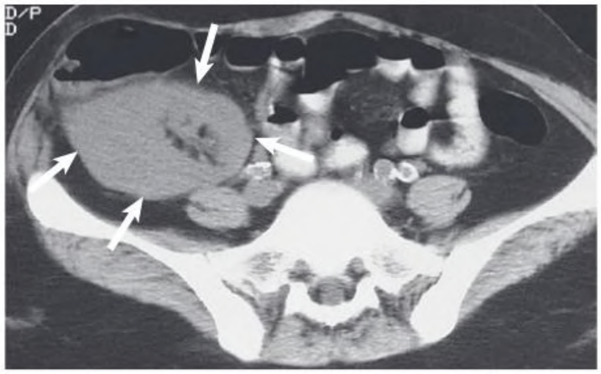

Fig. 6.16 Normal kidney transplant *(arrows)* on computed tomography.

COMPUTED TOMOGRAPHY

Computed tomography of the kidneys is performed to evaluate renal masses, locate ectopic kidneys (Figs. 6.15 and 6.16), investigate calculi, assess retroperitoneal masses, and evaluate the extent of parenchymal involvement in patients with acute pyelonephritis (Figs. 6.17 and 6.18). Helical CT scanners allow the abdomen and pelvis to be scanned at submillimeter intervals with single breath–held acquisitions, which eliminates motion artifact. Newer multidetector row CT results in multiple slices of information (64-slice and even 320-slice machines) being acquired simultaneously, allowing the entire abdomen and pelvis to be covered in very fast acquisition times of less than 30 seconds. Although the improved CT imaging usually comes at a price of increased radiation exposure to the patient, new reconstruction software with *iterative reconstruction* significantly reduces radiation exposure by as much as 30%. The CT data can be reconstructed in multiple planes and even 3D for improved anatomic visualization and localization.

Tissue Density

The Hounsfield unit (HU) scale is a measurement of relative densities determined by CT. Distilled water at standard pressure and temperature is defined as 0 HU; the radiodensity of air is defined as −1000 HU. All other tissue densities are derived from this measurement (Table 6.4). Tissues can vary in their exact HU measurements and will

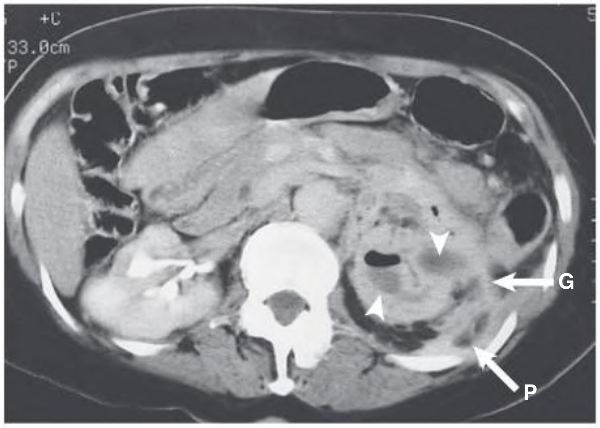

Fig. 6.17 Emphysematous Pyelonephritis. Contrast-enhanced computed tomography scan shows gas *(arrowheads)* within an enlarged left kidney and marked enhancement of the Gerota fascia *(G)* and posterior perirenal space *(P)*, indicative of inflammatory involvement.

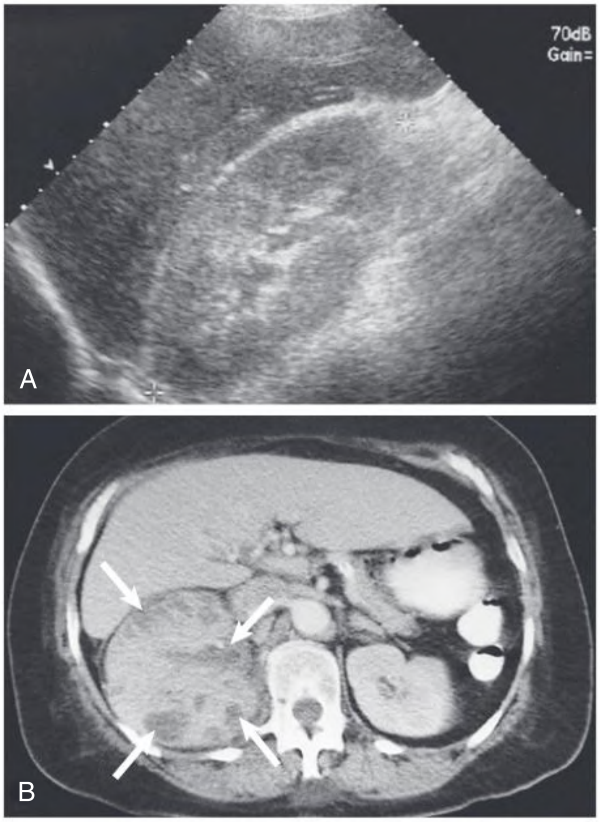

Fig. 6.18 Acute Pyelonephritis. (A) Ultrasound image demonstrates an enlarged echogenic kidney. Bipolar length of kidney is 12.9 cm. (B) Computed tomography scan with contrast enhancement obtained 24 hours later demonstrates multiple nonenhancing abscesses *(arrows).*

also change with contrast enhancement. Water, fat, and soft tissue often can look identical on the scan, depending on the window and level settings of the image, so actual HU measurement is essential to characterize the tissues accurately.

Contrast-Enhanced and Noncontrast Computed Tomography

CT examination of the kidneys can be performed with or without intravenous administration of contrast material. Noncontrast imaging allows the kidneys to be evaluated for the presence of calcium deposition and hemorrhage, which are obscured after contrast administration.

TABLE 6.4 Computed Tomography Determination of Density of Common Substances

Substance	Hounsfield Units[a]
Air	–1000
Fat	–120
Water	0
Muscle	+40
Bone	+400 or more

[a]The Hounsfield unit (HU) scale is a measurement of relative densities compared with distilled water.

Noncontrast CT is the examination of choice in patients with suspected nephrolithiasis and has replaced the KUB and IVU in most situations. The study consists of unenhanced images from the kidneys through the bladder for detection of calculi. CT is highly sensitive (97%–100%) and specific (94%–96%) for diagnosis of urinary calculi.[10] Noncontrast CT can identify a possible obstructing calculus and the extent of parenchymal and perinephric involvement.

In cases other than stone evaluation, the kidneys are imaged after contrast administration. The kidneys are imaged in the corticomedullary phase for evaluation of the renal vasculature and in the nephrographic phase for evaluation of the kidney parenchyma. The degree of enhancement can be assessed in both solid masses and complex cysts.

Computed Tomography Urography

Delayed images through the kidneys and bladder are performed for evaluation of the opacified and distended collecting system, ureters, and bladder.[11,12] After acquisition, the axial images can be reformatted into coronal or sagittal planes to optimize visualization of the entire collecting system (Fig. 6.19). The CT study can be tailored to the individual clinical scenario. For example, the corticomedullary phase can be eliminated to decrease the radiation dose if there is no concern about a vascular abnormality or no need for presurgical planning. A diuretic or saline bolus can be administered after contrast to better distend the collecting system and ureters during the excretory phase.

The kidneys should be similar in size and show equivalent enhancement and excretion. During the corticomedullary phase, there is brisk enhancement of the cortex. The cortical mantle should be intact. Any disruption of the cortical enhancement requires further evaluation; it may be caused by acute pyelonephritis (see Fig. 6.18), scarring, mass lesions, or infarction (Fig. 6.20). During the excretory phase, the entire kidney and renal pelvis enhance. Delayed excretion and delay in pelvicalyceal appearance of contrast material may be found in obstruction (Fig. 6.21) but also in parenchymal disease such as acute tubular necrosis (ATN).

The ureters are often seen segmentally because of active peristalsis. The ureters should be free of filling defects and smooth. In the abdomen, the ureters lie in the retroperitoneum, passing anterior to the transverse processes of the vertebral bodies. In the pelvis, the ureters course laterally and posteriorly, eventually draining into the posteriorly located vesicoureteral junction. At the vesicoureteral junction, the ureters gently taper. Medial bowing or displacement of the ureter is often abnormal and can be seen secondary to ureter displacement from retroperitoneal masses, lymphadenopathy, and retroperitoneal fibrosis.

The bladder should be rounded and smooth walled. Benign indentations on the bladder include the uterus, prostate gland, and bowel. In chronic bladder outlet obstruction and neurogenic bladder, numerous trabeculations and diverticula may be seen around the bladder outline. Other incidental bladder diverticula, such as the Hutch diverticulum,

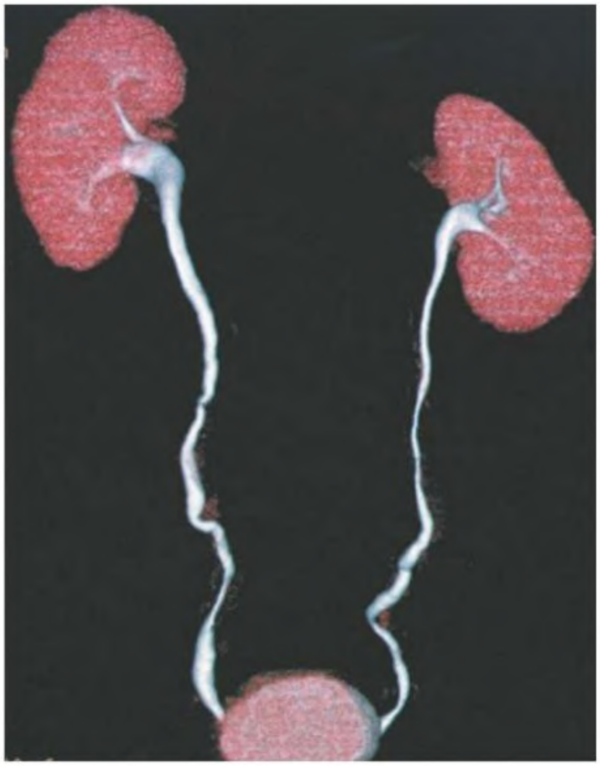

Fig. 6.19 **Noncontrast Computed Tomography (CT) of Bladder and Kidney.** Computer-reformatted, volume-rendered CT urogram obtained from axial CT acquisition.

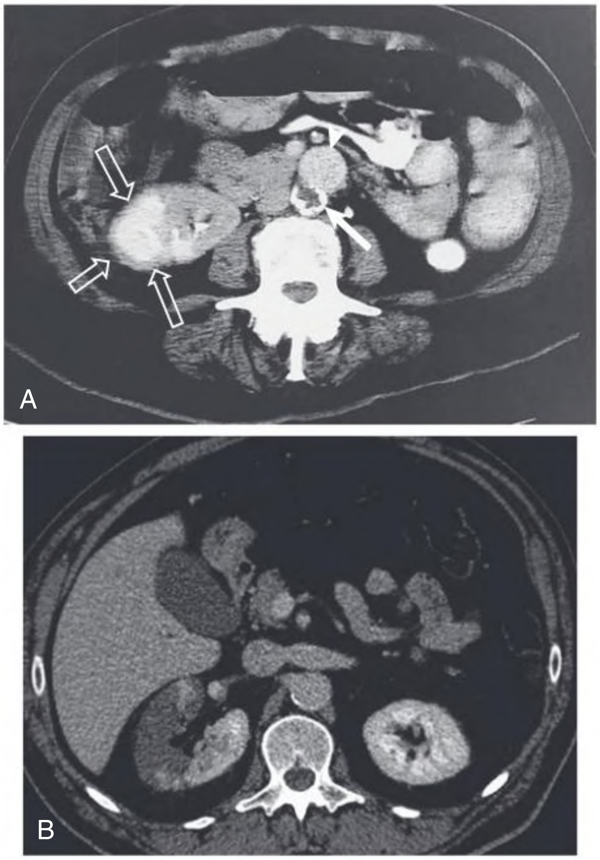

Fig. 6.20 **Renal Infarction.** (A) Computed tomography (CT) scan showing a chronic calcified infarct *(open arrows)* involving half of the right kidney after aortic bypass surgery. The native aorta has a densely calcified wall *(arrow)*. The aortic graft is anterior to the native aorta *(arrowhead)*. (B) CT scan showing acute infarct in the right kidney with decreased parenchymal enhancement *(arrows)*.

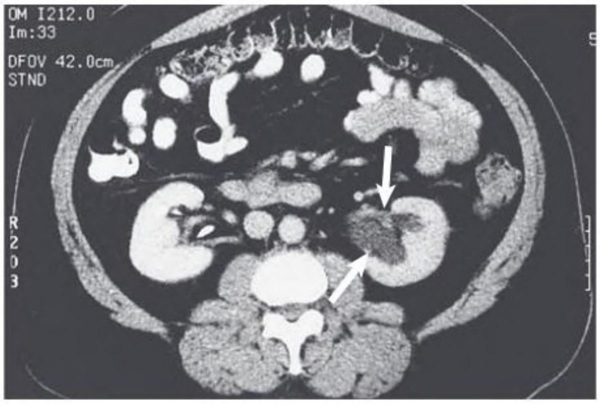

Fig. 6.21 **Delayed Excretion in Left Kidney Secondary to Distal Calculus.** Contrast-enhanced computed tomography scan shows dilated left renal pelvis *(arrows)*.

can be seen and are usually of no significance unless they are large and do not empty completely on voiding.

Computed Tomographic Angiography

Helical scanning facilitates CT angiography (CTA), which can produce images similar to conventional angiograms but is less invasive. A bolus of contrast material is administered, and the images are reconstructed at 0.5- to 3-mm consecutive intervals. The contrast bolus is timed for optimal enhancement of the aorta. The tightly focused and narrow CT beam allows higher resolution and better subsequent multiplanar reconstructions. The aorta and branch vessels are well demonstrated (Fig. 6.22). This technique is widely used in living transplant donor evaluation (see Fig. 104.2), providing information not only on arterial and venous anatomy but also on size, number, and location of the kidneys and any ureteral anomalies of number or position.

In addition, CTA can be used to evaluate for atheromatous renal artery stenosis, with sensitivity of 96% and specificity of 99% for the detection of hemodynamically significant stenosis compared with digital subtraction angiography (DSA).[2] Furthermore, CTA allows visualization of both the arterial wall and lumen, which helps in planning renal artery revascularization procedures. Another advantage of CTA is the depiction of accessory renal arteries and nonrenal causes of hypertension, such as adrenal masses. CTA can be used to diagnose fibromuscular dysplasia but has a lower sensitivity (87%) than DSA.[3]

Dual-Energy Computed Tomography

In dual-energy CT (DECT), two or more CT data sets are acquired using different tube potentials, usually 140 and 80 kilovolts (kV), resulting in different x-ray spectra. The density values at the acquired spectra differentiate materials on the basis of the photoelectric effect. The most common application of DECT is to allow the acquisition of "virtual noncontrast" images. A single acquisition after the administration of contrast can, using the photoelectric characteristics of iodine, be used to create a full set of images that appear as though no contrast was administered (Fig. 6.23A). These can be used to better identify kidney stones, differentiate stones and calcification from early excreted contrast in the urinary system, and can be compared against the post-contrast images to verify whether or not something is demonstrating enhancement. Among the more specialized uses of DECT include allowing kidney stone differentiation (i.e., differentiation of uric acid from magnesium or calcium), which may permit tailored treatment strategies.[4,5] Radiation dose in DECT has been shown to be the same as traditional CT, and optimized protocols can even reduce overall dose.[6]

Limitations of Computed Tomography

CT does have some limitations. The cradle that the patient lies on usually has an upper weight limit of 100 to 200 kg (300–400 lb), but newer scanners can now accommodate up to 270 kg (600 lb). Obese patients often have suboptimal scans because of weight artifact and need higher radiation exposures to adjust for x-ray attenuation. Contrast-enhanced CT studies may be contraindicated in patients with an allergy to radiographic contrast. For further discussion on contrast use in patients with impaired kidney function, see Radiologic Contrast Agents.

CT is very sensitive to metal artifact and patient motion. Retroperitoneal clips and intramedullary rods will cause extensive streak artifact, which severely degrades the images. However, recent advances in image reconstruction can be used to significantly decrease the streak artifact from metal implants, although this is still not perfect (see Fig. 6.23B). Patients who are unable to remain motionless will also have suboptimal or even nondiagnostic studies, and sedation or general anesthesia may be needed to obtain diagnostic scans, particularly in children. Critically ill patients may not be stable enough for transport to the CT suite, and ultrasound can be considered as an alternative to CT in that setting.

Fig. 6.22 **Normal Renal Arteries.** Three-dimensional reformatted computed tomography angiogram.

MAGNETIC RESONANCE IMAGING

Although it rarely should be the first examination used to evaluate the kidneys, MRI is typically an adjunct to other imaging. The major advantage of MRI over other modalities is direct multiplanar imaging. CT is limited to slice acquisition in the axial plane of the abdomen, and coronal and sagittal planes are acquired only by reconstruction, which can lead to loss of information.

Images on MR are typically described as T1 or T2 based on their acquisition. Fat is bright on T1 and not as bright on rapid acquisition T2 sequences (Fig. 6.24). The sequences and imaging planes selected must be tailored to the individual MR study. Diffusion-weighted imaging (DWI) evaluates the freedom of water molecules to diffuse in tissues; restriction of diffusion is imaged as bright areas on the DWI image set. Using the DWI data, regional apparent diffusion coefficients of the tissue can be calculated and an image of the distribution of the coefficients of the tissue can be produced. Dark areas on what is known as an apparent diffusion coefficient map are seen in infection, neoplasia, inflammation, and ischemia (Fig. 6.25). Because diffusion occurs in three dimensions, more advanced acquisitions can be obtained that show the direction of diffusion. This process is called diffusion tensor imaging (DTI). This is mostly seen in brain and neurologic imaging for imaging axonal tracts, but visualization of the orientation of renal tubules can be done with this method as well (Fig. 6.26).

On T1-weighted sequences, the normal kidney cortex is higher in signal than the medulla, producing a distinct corticomedullary differentiation, which becomes indistinct in parenchymal kidney disease. It is analogous to the echogenic kidney seen on ultrasound. On rapid acquisition T2 sequences, the corticomedullary distinction is not as

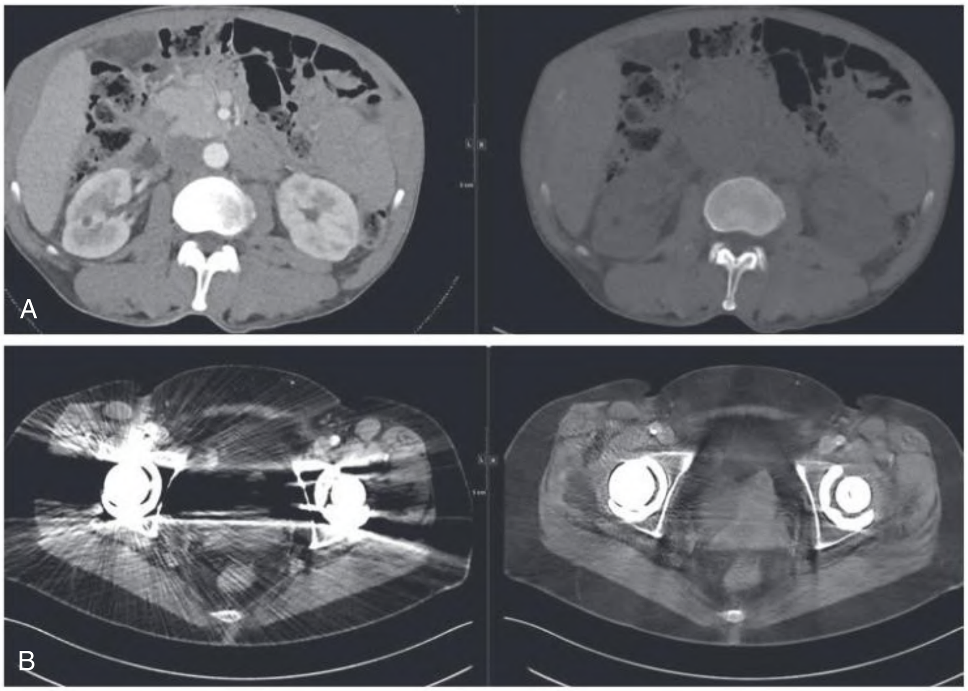

Fig. 6.23 (A) Following only one postcontrast computed tomography (CT) acquisition with dual energy, mathematical algorithms are used to obtain two sets of images. On the left is the standard postcontrast image, and on the right is the virtual noncontrast calculated image. (B) New CT software postprocessing permits much more diagnostic image quality with metal artifact reduction. The usual image obtained is at *right,* and the image using metal artifact reduction is at *left.*

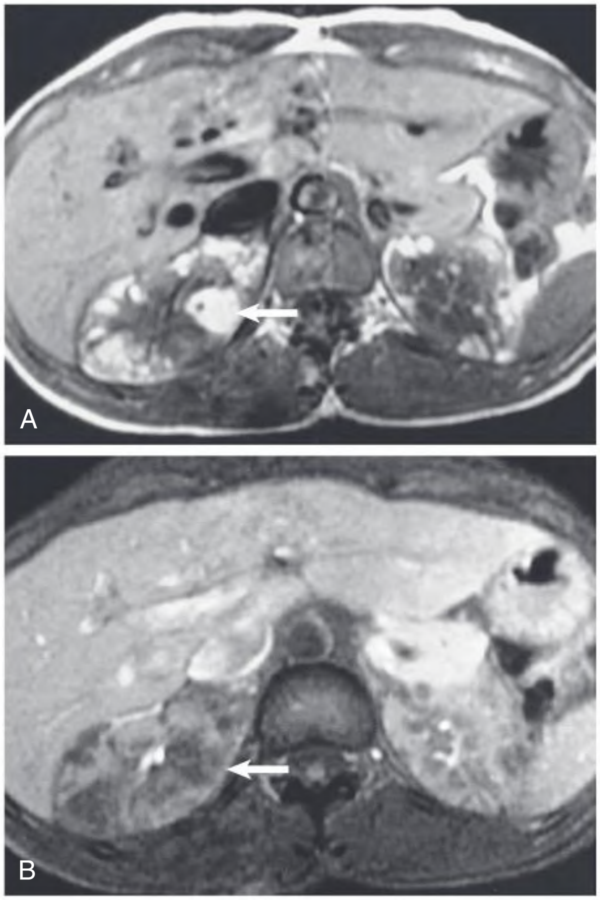

Fig. 6.24 **Tuberous Sclerosis on Magnetic Resonance (MR) Imaging.** Multiple renal angiomyolipomas are seen. (A) T1-weighted MR image. The tumors are high in signal on T1 because of their fat; *arrow* shows the largest tumor. (B) T1-weighted MR image with fat suppression. The fat within the tumors is now low in signal *(arrow).*

sharp but should still be present. Usually, at least one sequence is performed in the axial plane. Sagittal and coronal images cover the entire length of the kidney and can make some subtle renal parenchymal abnormalities more conspicuous (Fig. 6.27).

Contrast-Enhanced Magnetic Resonance Imaging

As with CT, intravenous contrast material can be administered to allow further characterization of kidney lesions. Gadolinium is a *paramagnetic* contrast agent frequently used in MRI. Nephrotoxicity to gadolinium agents is rare and insignificant.[7] A scleroderma-like condition could be seen with early contrast agents in subjects with chronic kidney disease (CKD), but this is not observed with agents used today (see Magnetic Resonance Contrast Agents). Paramagnetic contrast agents are being evaluated for measurement of glomerular function.

After injection of gadolinium, the vessels appear high in signal, or white, on T1-weighted sequences. Multiple images can be obtained in a single breath–held acquisition. This technique is useful for lesion characterization in patients who cannot receive iodinated contrast material. As with contrast-enhanced CT, the kidneys initially show symmetric cortical enhancement, which progresses to excretion. A delay in enhancement can be seen with renal artery stenosis.

Magnetic Resonance Urography

Two techniques are used to perform magnetic resonance urography (MRU).[13] The first technique is sometimes called *static* MRU. Because

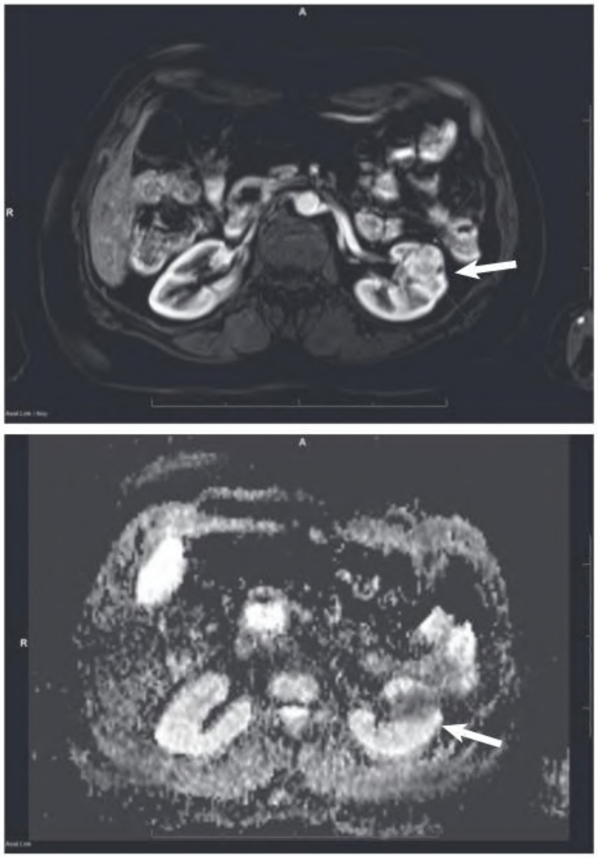

Fig. 6.25 **Magnetic Resonance Image of Kidney Tumor.** T1-weighted, contrast-enhanced image shows enhancing left renal tumor.

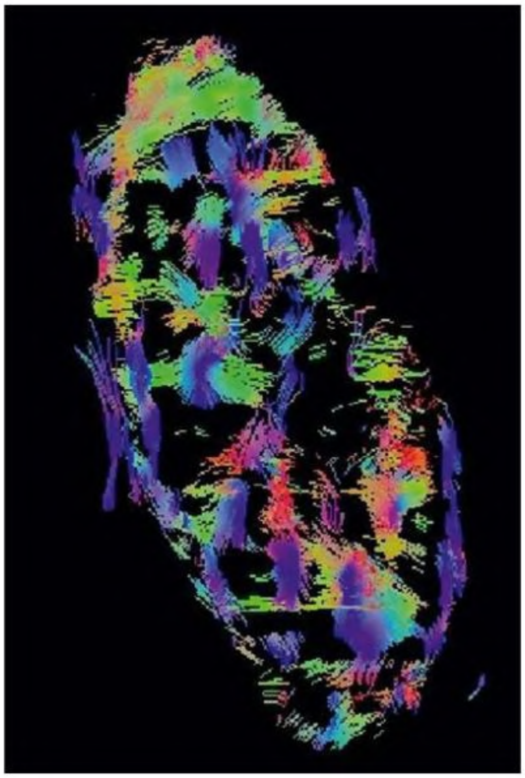

Fig. 6.26 **Diffusion Tensor Imaging (DTI) Permits Imaging of Eigenvectors of Relative Diffusion Through Tissues.** This image shows a kidney DTI with visualization of the diffusion through the medullary tubules.

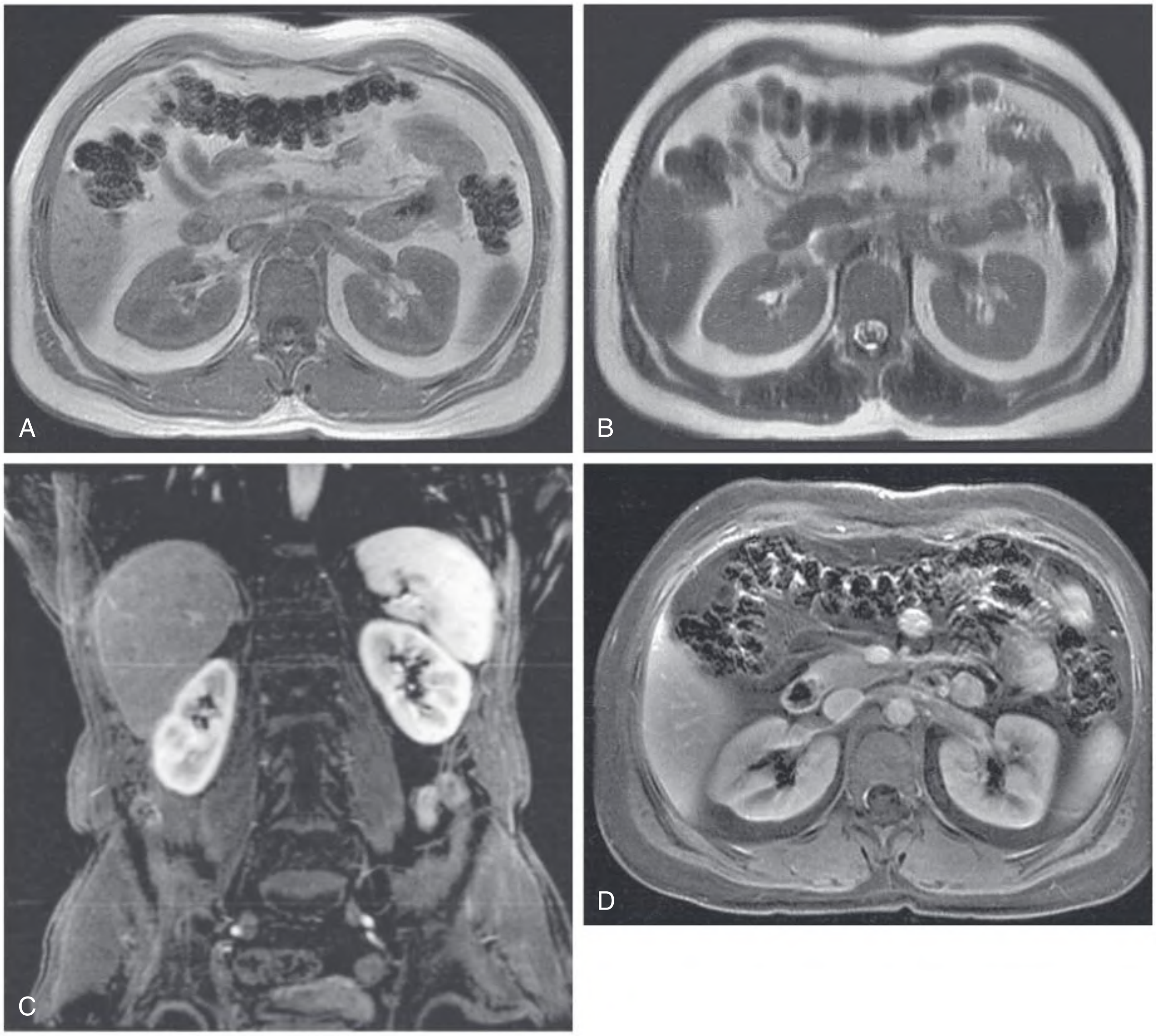

Fig. 6.27 Normal Magnetic Resonance (MR) Images Through Kidneys. (A) T1-weighted MR image. Note the distinct corticomedullary differentiation. (B) Fast spin echo MR image. Urine in the collecting tubules causes the high signal within the renal pelvis on this sequence. (C) Coronal T1-weighted, fat-suppressed MR image after contrast administration. (D) Axial T1-weighted, fat-suppressed image after contrast administration.

urine contains abundant water, it will demonstrate high signal on a T2-weighted image. Therefore, a heavily T2-weighted sequence accentuates the static fluid in the collecting system and ureters, which stands out against the darker background soft tissues. Static MRU can be performed rapidly, which is a benefit in imaging of children. A disadvantage is that any fluid in the abdomen or pelvis, such as fluid collections or fluid in the small bowel, will demonstrate a similar bright signal that can obscure superimposed structures. Also, the collecting system and ureters need to be distended for acquisition of good MR images.

The second technique, often referred to as *excretory* MRU, is similar to CT urography (CTU). Intravenous administration of gadolinium is followed by T1-weighted imaging. This technique allows some assessment of kidney function because the contrast is filtered by the kidney and excreted into the urine (see Fig. 58.10). The opacified collecting system and ureters are well seen, and a diuretic can be administered to further dilate the renal pelvis and ureters if necessary. MRU has limited capacity to detect calculi because calcification is poorly visualized by MRI.

Because CTU and MRU are comparable for identifying the cause and anatomic location of urinary obstruction, the choice of modality is a matter of local preference. CTU is the better choice in the evaluation of urinary tract calculi. In patients with reduced glomerular filtration rate (GFR) caused by obstruction, MRU is superior to CTU in identifying noncalculous causes of obstruction, whereas CTU is superior in identifying calculi as a cause of obstruction.[14] CTU is also more widely available, faster, and less expensive than MRU. MRU is better suited in patients with allergy to iodinated contrast agents and sometimes in children when radiation is an issue. MRU is also useful in depicting the anatomy in patients with urinary diversion to bowel conduits.

Magnetic Resonance Angiography

Although magnetic resonance angiography (MRA) can be performed with or without intravenous contrast, contrast provides better images. The aorta and branch vessels are beautifully demonstrated (Fig. 6.28). By adjustment of timing and type of sequences, the abdominal venous structures can be visualized (Fig. 6.29). MRA is performed to evaluate the renal arteries for stenosis and is less invasive than catheter angiography

Fig. 6.28 **Magnetic Resonance Angiography.** Coronal three-dimensional magnetic resonance angiogram after contrast administration shows normal renal arteries.

Fig. 6.29 Magnetic resonance venography.

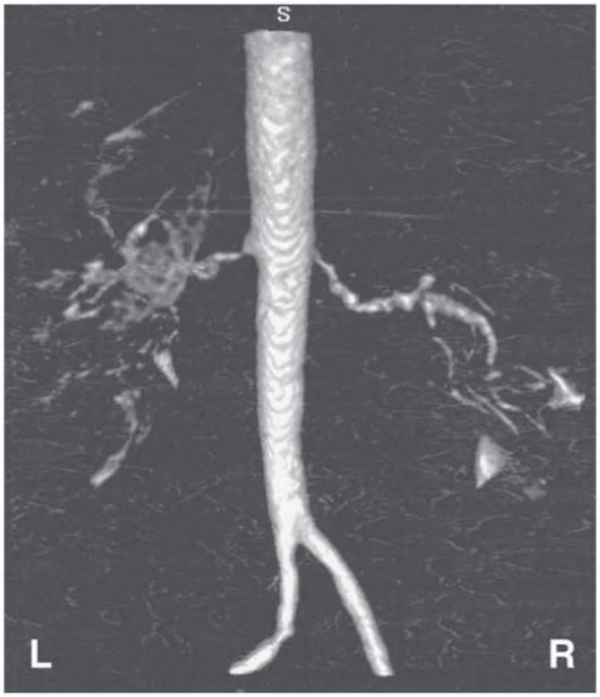

Fig. 6.30 **Dysplasia on Magnetic Resonance (MR) Angiography.** Posterior view of coronal three-dimensional MR angiogram shows typical beaded appearance of fibromuscular dysplasia of the right renal artery and focal high grade stenosis of the origin of the left renal artery.

(Fig. 6.30). Technical advances, including faster sequences, now give sensitivity of 97% and specificity of 93% compared with DSA for contrast-enhanced MRA in the detection of renal artery stenosis.[15] MRA without gadolinium has a lower sensitivity (53%–100%) and specificity (65%–97%) for detection of renal artery stenosis.[16] MRA has limited ability to assess accessory renal arteries and therefore is not an ideal study to evaluate fibromuscular dysplasia. It has become the most commonly used modality in patients with hypertension, declining kidney function, or allergy to iodinated contrast agents.[17] Where MRA is unavailable, Doppler ultrasound can be used.

Disadvantages of Magnetic Resonance Imaging

Like CT, MRI has some disadvantages. The table and gantry are confining, so claustrophobic patients may be unable to cooperate. Patients with some types of internal metallic hardware cannot undergo MRI. MRI safety guidelines have been developed with an extensive list of devices that are or are not MRI approved, and any devices in the patient need to be checked against this list. Examples of contraindications include neural stimulator devices and cerebral aneurysm clips. Many cardiac pacemakers are now compatible with MRI, but if the patient is pacemaker dependent, they cannot be scanned.

Determination of in-stent stenosis is impossible, as metallic artifacts from renal artery stents completely obscure the lumen. Even with the new, fast imaging techniques, patients need to be able to cooperate with breath-holding instructions to minimize motion-related artifacts. The safety of the use of MRI contrast agents in patients with impaired kidney function is discussed later (see Magnetic Resonance Contrast Agents).

MRI can be used in the intensive care unit and in critically ill patients only if they are stable enough to be transported to the MRI suite and have no implanted metallic devices. Ventilated patients can undergo MRI; however, specific MRI-compatible, nonferromagnetic ventilators and other life support devices must be used. Because of the confined nature of the MRI gantry, visualization and monitoring of the patient during the scan are compromised.

Incidental Findings

Incidental kidney lesions are being found with increasing frequency. Almost 70% of renal cell carcinomas are discovered incidentally on imaging studies performed for other reasons. There is an age-dependent incidence of renal cysts, from about 5% in patients younger than 30 years to almost one-third of those older than 60 years.[18] The differentiation of solid and cystic lesions is the first mandate because as many as two-thirds of solid lesions are found to be malignant.[19] MRI is ideally suited for lesion evaluation and is often better than ultrasound, particularly for complex cystic lesions. Parameters being characterized include solid versus cystic appearance, overall lesion complexity, lesion enhancement, involvement of renal vasculature and collecting system, and extension into perirenal tissues and organs. Diffusion-weighted MRI sequences are often useful as a means of further differentiating benign and malignant solid lesions.

MEASUREMENT OF GLOMERULAR FILTRATION RATE

Renal blood flow and split kidney function can be evaluated by CT and MRI.[20-22] The attenuation of the accumulated contrast material within the kidney is directly proportional to the GFR. Taking into account the renal volume, the function of each kidney can be determined. Although both modalities yield similar information, MRI is used more in children and in patients with allergy to contrast agents. However, scintigraphy is still widely used for assessing kidney function, as discussed later.

ANGIOGRAPHY

Angiography is now most often performed for therapeutic intervention, such as embolotherapy or angioplasty and stenting, preceded by diagnostic angiography to evaluate the renal arteries for possible stenosis (Fig. 6.31). With improved resolution and scanning techniques, CTA and MRA have replaced conventional angiography, even for detection of accessory renal arteries, which are often small and bilateral but a possible cause of hypertension. However, angiography remains the gold standard reference test for the diagnosis of renal artery stenosis and fibromuscular dysplasia. There also remains a role for diagnostic angiography in the evaluation of medium- and large-vessel vasculitis and detection of renal infarction.

The conventional angiogram is performed through arterial puncture, followed by catheter placement in the aorta and subsequent selective renal artery catheterization as necessary. Contrast is administered intraarterially, and the images are obtained with DSA. DSA uses computer reconstruction and manipulation to generate the images, with the advantage that previously administered and excreted contrast material and bones can be digitally removed to better visualize the renal vasculature. Angiography can cause contrast-induced nephropathy and cholesterol embolization (see Chapter 41). Pathologic evidence of cholesterol embolization is common, but clinically significant symptoms occur infrequently (1%–2%).[23]

RENAL VENOGRAPHY

Catheter venography was once used for evaluation of renal vein and gonadal vein thrombosis and for renal vein sampling to measure renin, but it has largely been replaced with Doppler ultrasound, followed by contrast-enhanced CT or MRI (see Fig. 6.17).

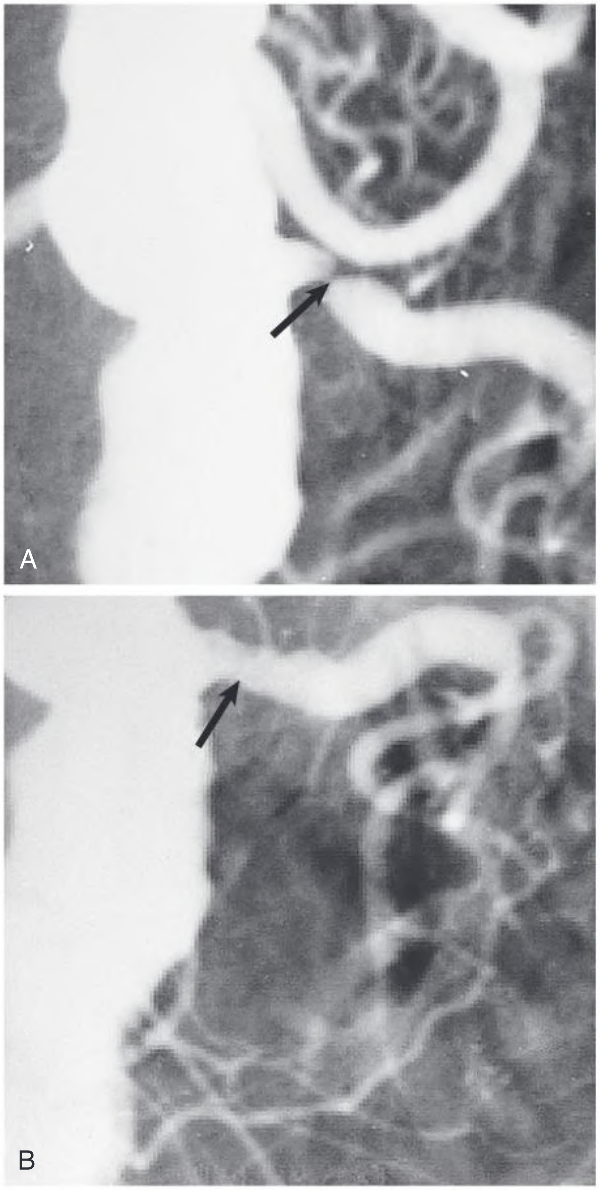

Fig. 6.31 **Left Renal Artery Stenosis and Angioplasty.** (A) Aortogram demonstrating a tight left renal artery stenosis *(arrow).* (B) Postangioplasty image with marked improvement of the stenosis *(arrow).* (Courtesy Dr. Harold Mitty.)

NUCLEAR SCINTIGRAPHY

Nuclear scintigraphy evaluates function and anatomy seen with other diagnostic imaging modalities. Radiotracers are designed to accumulate in tissues or organs on the basis of underlying functions unique to that organ. The gamma camera captures the photons from a radiotracer within the patient and generates an image. Single-photon emission computed tomography (SPECT) is a specialized type of imaging in which the emitted photons are measured at multiple angles, similar to CT, and multiplanar or even 3D images can be created. Three categories of radiotracers that differ in mode of renal clearance are used in kidney imaging: glomerular filtration, tubular secretion, and tubular retention agents (Table 6.5).

TABLE 6.5 Choice of Radionuclide in Kidney Imaging

Imaging Target	Radiotracer
Glomerular filtration rate	^{99m}Tc-DTPA
Glomerular filtration rate with renal impairment	^{99m}Tc-MAG3, ^{131}I-OIH
Effective renal plasma flow	^{99m}Tc-MAG3, ^{131}I-OIH
Kidney scarring	^{99m}Tc-DMSA, ^{99}Tc-GH
Renal pseudotumor	^{99m}Tc-DMSA
Upper renal tract obstruction	^{99m}Tc-DTPA
Upper renal tract obstruction with reduced GFR	^{99m}Tc-MAG3

^{99m}Tc-DMSA, Technetium 99m–labeled dimercaptosuccinate; *^{99m}Tc-DTPA,* technetium 99m–labeled diethylenetriaminepentaacetic acid; *^{99m}Tc-MAG3,* mercaptoacetyltriglycine; *^{99m}Tc-GH,* technetium 99m glucoheptonate; *^{131}I-OIH,* 131I orthoiodohippurate.

Scintigraphy remains superior to the other imaging modalities in the evaluation of renal flow. It is the study of choice in the evaluation of renal transplants and functional obstruction, especially when ultrasound evidence is equivocal. Scintigraphy is also widely used to measure GFR, although CT or MRI is preferred in some centers.

Although CT, MRI, and contrast-enhanced ultrasound can be used to evaluate kidney function (e.g., after nephron-sparing surgery), scintigraphy remains the preferred modality. Both CTA and MRA have replaced nuclear scintigraphy in the evaluation of renal artery stenosis and benign renal masses, such as a column of Bertin. Nuclear medicine is still used to assess the functional significance of renal artery stenosis independent of anatomy.

Glomerular Filtration Agents

Glomerular filtration agents can be used to measure GFR. Technetium 99m–labeled diethylenetriaminepentaacetic acid (^{99m}Tc-DTPA) is the most common glomerular agent used for imaging and also can be used for GFR calculation. In patients with poor kidney function, renal imaging with tubular secretion agents such as mercaptoacetyltriglycine (^{99m}Tc-MAG3) is superior to DTPA.[24,25]

Tubular Secretion Agents

^{99m}Tc-labeled MAG3 is handled primarily by tubular secretion and can be used to estimate effective renal plasma flow. The clearance rate for ^{99m}Tc-MAG3 is 340 mL/min.[26]

Tubular Retention Agents

Tubular retention agents include ^{99m}Tc-labeled dimercaptosuccinate (DMSA) and less often ^{99m}Tc-labeled glucoheptonate (GH). These agents provide excellent cortical imaging and can be used in suspected renal scarring or infarction, in pyelonephritis, and for clarification of renal pseudotumors. These agents bind with high affinity to sulfhydryl groups on the surface of proximal tubular cells.

Renogram

A renogram (or renal scintigram) is generated by scintigraphy and provides information about blood flow, renal uptake, and excretion. Time-activity graphs are produced that plot blood flow of the radiotracer into each kidney relative to the aorta. Peak cortical enhancement and pelvicalyceal clearance of the tracer are also plotted. DTPA or MAG3 can be used to generate the renogram. The relative radiotracer uptake can be measured and can provide split or differential information about kidney function (Fig. 6.32).

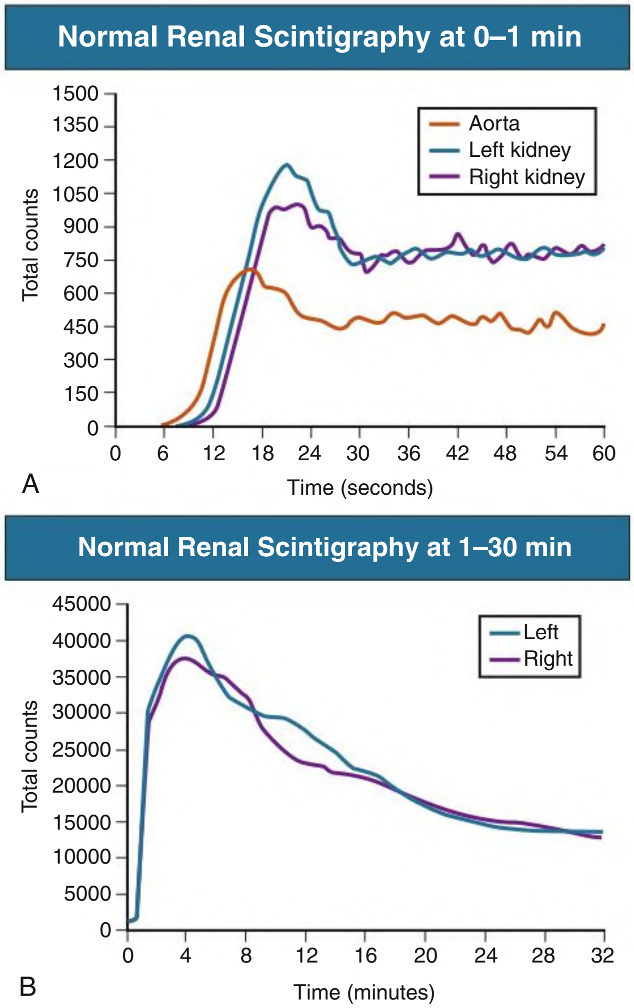

Fig. 6.32 Normal ^{99m}Tc-Labeled DTPA Study: Time-Activity Curves. (A) Early (0–1 minute), showing renal blood flow. (B) Later (1–30 minutes), showing renal uptake and excretion of tracer. (Courtesy Dr. Chun Kim.)

The blood pool or flow images are obtained after bolus injection of the radiotracers. Images are obtained with the gamma camera every few seconds for the first minute. The second component of the renogram evaluates kidney function by measuring radiotracer uptake and excretion by the kidney. In health, the peak renal cortical concentration occurs between 3 and 5 minutes after injection of tracer. Delayed transit of the isotope secondary to kidney dysfunction (e.g., ATN or rejection) or obstructive uropathy will alter the curve of the renogram.

In cases of suspected obstructive uropathy, a diuresis renogram can be obtained. A loop diuretic is injected intravenously when radiotracer activity is present in the renal pelvis; a computer-generated washout curve is obtained. In patients with true obstruction, activity will remain in the renal pelvis, whereas it will quickly wash out in patients without an obstruction (Fig. 6.33; see also Fig. 58.12).

Cortical Imaging

Imaging of the kidney cortex is performed with tubular retention agents, usually ^{99m}Tc-DMSA. Information about renal size, location, and contour can be obtained (Fig. 6.34). The cortical study is used most frequently for evaluation of renal scarring, particularly in children with reflux or chronic infections (see Chapter 61). Cortical imaging

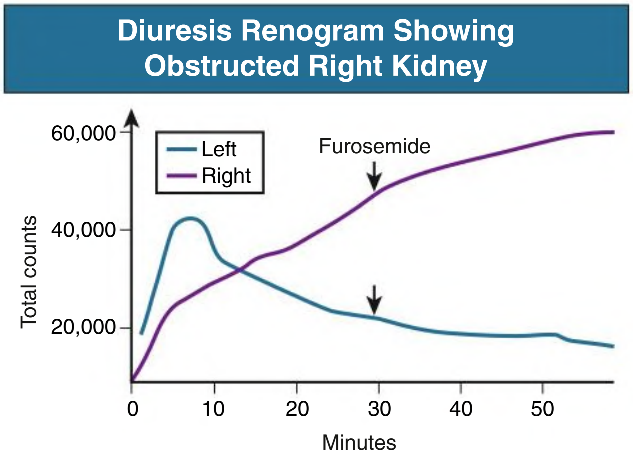

Fig. 6.33 Diuresis Renogram Showing Obstructed Right Kidney. Isotope continues to accumulate in the right kidney despite intravenous furosemide (given at 30 minutes). Isotope excretion in the left kidney is normal.

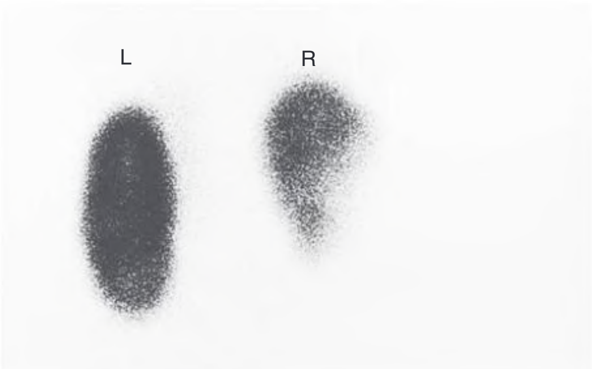

Fig. 6.34 Renal Infarct. ^{99m}Tc-DMSA scan in a newborn with an infarct of the right lower pole *(R)* secondary to embolus from umbilical catheter. (Courtesy Dr. Chun Kim.)

may be better than ultrasound in the evaluation of the young patient with urinary tract infection.[27] An infection, scar, or space-occupying lesion (tumor or cyst) will create a cortical defect, and correlation of the cortical defect site with other cross-sectional imaging should be performed to differentiate these entities.

Vesicoureteral Reflux

In children with suspected vesicoureteral reflux, a standard cystogram is obtained. If reflux is shown, follow-up is subsequently performed with radioisotope cystography, which exposes the child to a lower radiation dose and can be used to quantitate the bladder capacity when reflux occurs. The study is performed after instillation of technetium pertechnetate through a catheter into the bladder. Images are obtained during voiding.

Kidney Transplant

Kidney transplants are easily evaluated with scintigraphy. ^{99m}Tc-MAG3 is cleared through tubular secretion, which is maintained in most kidneys to a better degree than glomerular filtration in kidney failure. Because many transplant recipients have declining kidney function, ^{99m}Tc-MAG3 is the first-choice nuclide.

As with the normal kidneys, information about blood flow and function can be determined. Postoperative complications involving the artery, vein, or ureter are also well delineated. Nuclear imaging can help define ATN versus rejection in transplant patients with declining kidney function. Ultrasound with Doppler evaluation of resistive index is often a complementary investigation, and choice of imaging modality in part depends on local expertise and preference.

Fig. 6.35 Normal Positron Emission Tomography Scan. Note normal radiotracer uptake in brain, heart, intestines, and liver, with normal excretion in kidneys.

POSITRON EMISSION TOMOGRAPHY

Positron emission tomography (PET) scanning uses radioactive positron emitters, most often fluorine-18–labeled fluorodeoxyglucose (FDG). The FDG is intravenously injected and distributes in the body according to metabolic activity. Any process, such as a tumor or infection, that causes increased metabolic activity will result in an area of increased uptake on the scan. These areas of abnormality need to be differentiated from normally hypermetabolic tissues, such as brain, liver, bone marrow, and to some extent heart and bowel (Fig. 6.35). Because FDG is cleared through the kidneys and excreted in the urine, which can obscure renal masses or infection, PET scanning has a limited role in renal imaging but is useful in the staging and follow-up of metastatic renal cancer.[28,29] Other ligands exist with lower levels of urinary excretion, but the application of this in renal imaging is still being investigated.

MOLECULAR IMAGING

With molecular imaging, radiology is moving from the identification of generic anatomy and nonspecific enhancement patterns to assessment of specific molecular differences in tissues and disease processes. Nuclear imaging presently is molecular based but still nonspecific (e.g., FDG-PET, renal DTPA). The newer focus of molecular imaging studies is dynamic processes such as metabolic activity, cell proliferation, apoptosis, receptor status, fibrosis, and antigen modulation.

Typically, this involves imaging of biochemical and physiologic processes. Techniques are being developed with optical scanning, MRI, and ultrasound as well as with radionuclides.

Applications are established in clinical practice, particularly in oncology (e.g., CD20 imaging in lymphoma), and work is under way for renal-specific molecular imaging. For example, MR renal cell imaging may be available soon to help differentiate ATN from renal rejection and renal cell cancer from benign tumors.

RADIOLOGIC CONTRAST AGENTS

X-Ray Contrast Agents

Contrast agents continue to have a role in many imaging techniques. A triiodinated benzene ring forms the chemical basis for CT intravascular contrast agents. Conventional contrast agents have high osmolality, about five times greater than plasma osmolality. Modifications to the benzene ring have led to newer contrast agents, including low-osmolar (which is still hyperosmolar compared with normal plasma) and more recently iso-osmolar nonionic agents, which are less nephrotoxic.

In patients with normal GFR, the kidneys eliminate almost all the contrast agent. Extrarenal routes of excretion include the liver and bowel wall and account for less than 1% of elimination, but this can increase when kidney function is compromised. The half-time in patients with normal GFR is 1 to 2 hours, compared with 2 to 4 hours in dialysis patients.[30]

Contrast reactions for iodinated agents occur in 3.1% to 4.7% of patients.[31-33] Of those patients who have a contrast reaction, 20% will experience a reaction on reexposure that may be similar or worse. Contrast reactions can be anaphylactoid or chemotoxic reactions. The anaphylactoid reactions mimic an allergic response, whereas the chemotoxic reactions are believed to be mediated by direct toxic effects of the contrast material. The exact mechanism of contrast reaction is not known but is likely to be multifactorial. Formation of antigen-antibody complexes, complement activation, protein binding, and histamine release have been cited as possible mechanisms.

Reactions may be minor, intermediate, or severe. Minor reactions include heat sensation, nausea, and mild urticaria. Intermediate reactions include vasovagal reaction, bronchospasm, and generalized urticaria. Severe reactions include profound hypotension, pulmonary edema, and cardiac arrest. The use of low-osmolar or iso-osmolar contrast agents reduces the incidence of minor and intermediate contrast reactions. The reported incidence of death related to high-osmolar contrast agents is 1 in 40,000. Immediate treatment of reactions should be directed toward the symptoms. In patients with a history of contrast allergy, pretreatment is recommended with antihistamines and corticosteroids on reexposure.

Contrast-Induced Nephropathy

Although acute kidney injury (AKI) associated with the administration of contrast material has been reported as the third most common cause of in-hospital AKI, recent data indicate this risk has been markedly overestimated.[33] The original studies and reports often occurred using high-osmolarity contrast agents in subjects who were not well hydrated. Biopsies from those patients often showed osmotic nephrosis or acute tubular necrosis.[34] Today, contrast-induced AKI is rare, even in patients with GFR less than 45 mL/min/1.73m^2, and thus the risk of AKI does not preclude the use of intravenous contrast if clinically necessary. Despite the concern for long-term complications, there is evidence suggesting that intravenous contrast carries little risk for dialysis dependence or mortality.[35]

Studies of the pathogenesis of contrast-induced nephropathy have suggested that the injury is due to the high osmolality of the agent, which can induce renal vasoconstriction, activation of the polyol-fructokinase pathway in the proximal tubule associated with oxidative stress,[36] and possibly the effects of the acute osmotic-induced uricosuria. Most underlying cellular events were thought to occur within the first 60 minutes after administration of the contrast agent, with the greatest risk in the first 10 minutes. Tubular injury produces oxygen free radicals, possibly from the vasoconstriction. In animal studies, reduction in antioxidant enzymes associated with hypovolemia contributes to the injury.[37]

An important differential diagnosis for contrast-induced nephropathy in patients with vascular disease undergoing catheter angiography is cholesterol embolization (see Chapter 41).

Possible risk factors include preexisting chronic kidney disease, diabetes, cardiovascular disease, use of diuretics, advanced age (>75 years), multiple myeloma in dehydrated patients, hypertension, uricosuria, and high-dose contrast. In patients with kidney failure, contrast administration may result in fluid overload because of thirst induced by the osmotic load. Hydration with normal saline is the mainstay of prevention; there is no substantial evidence that sodium bicarbonate offers any advantage over saline.[38] Oral *N*-acetylcysteine, a thiol-containing antioxidant, is often given in conjunction with hydration but has not proved consistently to be protective.[39] Note that even the use of hydration is controversial, with excessive hydration putting these patients at risk of fluid overload.[40]

In patients with estimated GFR less than 60 mL/min/1.73m^2, low-osmolar or iso-osmolar contrast agents can be used and the doses reduced. Repetitive, closely performed contrast studies should be avoided. In high-risk patients, alternative imaging studies (ultrasound, MRI, or noncontrast CT) always should be considered. Issues related to contrast-induced nephropathy are further discussed in Chapter 72.

Magnetic Resonance Contrast Agents

The two classes of MRI contrast agents are diffusion and nondiffusion agents. Diffusion agents, with appropriate timing of imaging sequences, can delineate vessels and parenchymal tissues. Nondiffusion agents remain in the bloodstream and are primarily useful for MRA. All the contrast agents are based on the paramagnetic properties of gadolinium. Gadolinium itself is highly toxic and is given only when it is tightly chelated (e.g., Gd-tetraazacyclododecane-1,4,7,10-tetraacetic acid [Gd-DOTA], Gd-diethylenetriamine penta-acetic acid [Gd-DTPA]).

Minor reactions such as headache and nausea occur in 3% to 5% of patients, but life-threatening reactions and nephrotoxic reactions are rare. In patients with reduced GFR, a rare severe reaction (nephrogenic systemic fibrosis [NSF]), has been described (see Chapter 91). Guidelines confirm that MRI using high-risk gadolinium-containing contrast agents is contraindicated in patients with AKI and in those with chronic kidney disease stages 4 and 5 (i.e., GFR <30 mL/min/1.73 m^2),[16] because some agents have been linked to NSF. Cases of NSF are particularly associated with the linear structure chelates (such as gadodiamide, gadopentetate, and gadoversetamide). However, some linear chelates have a higher dissociation constant and to date have not been associated with NSF (such as gadobenate).[41] The newer macrocyclic gadolinium-containing contrast agents (e.g., gadoterate and gadoteridol) also have not been associated with NSF, even in patients with GFR less than 30 mL/min/1.73 m^2. Recently, studies have shown gadolinium deposition and retention in tissues, especially the brain, and especially after multiple injections of the contrast material. To date, there are no documented adverse events associated with this retention. The choice of gadolinium agent in each case should be determined by discussion between the nephrologist and radiologist.

Radiomics and Artificial Intelligence

Radiomics and artificial intelligence are areas of ongoing research. Radiomics deals with the conversion of images into data formats that allow assessment and quantification of pixel or voxel gray levels (spatial heterogenicity). This allows texture analysis that can be used as is or with further machine learning and adaption of artificial intelligence algorithms.

These tools are being studied for differentiating benign and malignant lesions as well as characterizing renal cell cancer types. Additionally, these tools may be useful in following response of renal cell cancer to chemotherapeutic agents.[42-44]

Although these techniques show promise, there are still many limitations and challenges before these tools can be adapted to routine clinical use.

SELF-ASSESSMENT QUESTIONS

1. On ultrasound, compared with the liver, the normal adult kidney cortex is:
   A. isoechoic.
   B. hyperechoic.
   C. hypoechoic.
   D. anechoic.
2. Which of the following imaging modalities has the *least* chance of inducing contrast nephropathy?
   A. CO_2 angiography.
   B. Contrast-enhanced CT.
   C. Contrast-enhanced MRI.
   D. Intravenous urography.
3. The imaging modality that exposes the patient to the *least* radiation risk is:
   A. CT urography.
   B. dual-energy CT with virtual noncontrast imaging.
   C. intravenous urography.
   D. diffusion-weighted MRI.
4. Which of the following is a mandatory reason that an MRI of the kidneys *cannot* be performed?
   A. Cardiac pacemaker in a patient who is not pacemaker dependent.
   B. Titanium total hip replacement less than 6 weeks old.
   C. When eGFR is 35 mL/min.
   D. History of cerebral aneurysm clip.
5. Which of the following is usually the recommended *best* modality to evaluate for kidney stones?
   A. Noncontrast CT scan.
   B. MRI.
   C. Ultrasound.
   D. Nuclear medicine renogram.

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